

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The circular economy, biocapacity & ecological footprints - Set 1

Mark scheme

AQA spec 3.6.4; Sustainability; textbook pp. 374-389

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Cycling of materials (like biogeochemical cycles) - recycling, re-use (1) • Energy from renewable sources (1) • Human activities should support ecosystems / conserve biodiversity (1) • Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis) (1) • Diverse systems are more resistant to change (1) • Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling (1)	6
01.2	<i>1 mark for each point.</i> • Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae (1) • Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs (1)	2
01.3	<i>1 mark for each point.</i> • Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them (1) • A common approach gives a level playing field, coordinated monitoring and support for less successful countries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
02.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
02.2	<i>1 mark.</i> • Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	
03.1	<i>1 mark for each point.</i> • Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced (1) • Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares (1)	2

Question	Marking guidance	Marks
03.2	<i>1 mark for each point.</i> • A hectare of land / sea with the world-average biological productivity (1) • Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared (1)	2
03.3	<i>1 mark for the correct answer.</i> • $8.1 / 1.7 = 4.8$ (accept 4.7-4.8) (1)	1
03.4	<i>1 mark for each point.</i> • Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint (1) • Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO₂ accumulates (overshoot) (1) • Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016) (1)	3
03.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Renewable energy and energy conservation (carbon footprint is the largest component) (1) • Reduced meat consumption / food waste; local seasonal food (1) • Circular economy: repair, re-use, recycling; longer product lives (1) • Public transport, efficient buildings (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	
04.1	<i>1 mark for each point.</i> • All greenhouse gases expressed as CO₂ equivalent (allowing for different warming strengths) (1) • Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) (1) • A fair share of the infrastructure: roads, car parks, signals, street lighting (1)	3
04.2	<i>1 mark for each point.</i> • Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels (1) • Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time (1)	2
04.3	<i>1 mark for each point.</i> • Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too (1) • Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth (1) • Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated (1) • Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected (1) • Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers (1)	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	

Question	Marking guidance	Marks
05.1	<i>1 mark for each point.</i> • Diversity: HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change (1) • No single source could meet demand all year; battery storage on the island grid smooths fluctuations (1) • Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) (1)	3
05.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife (1) • Wood biomass heating; insulation such as straw-bale housing; some electric vehicles (1) • Habitat protection and creation: fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow); eagles and hen harriers nesting (1) • Recyclables shipped off; garden / food waste composted; low-impact food production; beach litter collection (1)	3
05.3	<i>1 mark for each point.</i> • Small groups can organise, agree and act quickly; decisions benefit people who see the results directly (1) • Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away (1)	2
05.4	<i>1 mark for each point.</i> • Renewable energy is abundant but geographically and temporally variable; a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand (1) • DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	